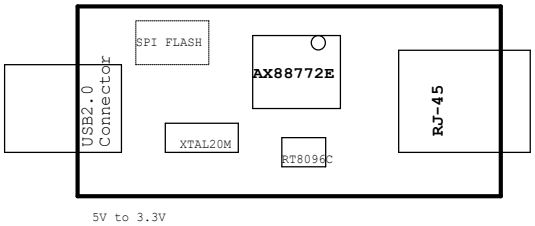
AX88772E Demo Board Schematic

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System Block (This Page)

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AX88772E
SPI Flash
I2C EEPROM
20MHz Crystal
USB/RJ-45 Connectors
Power Circuit
RT8096C

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Revision History

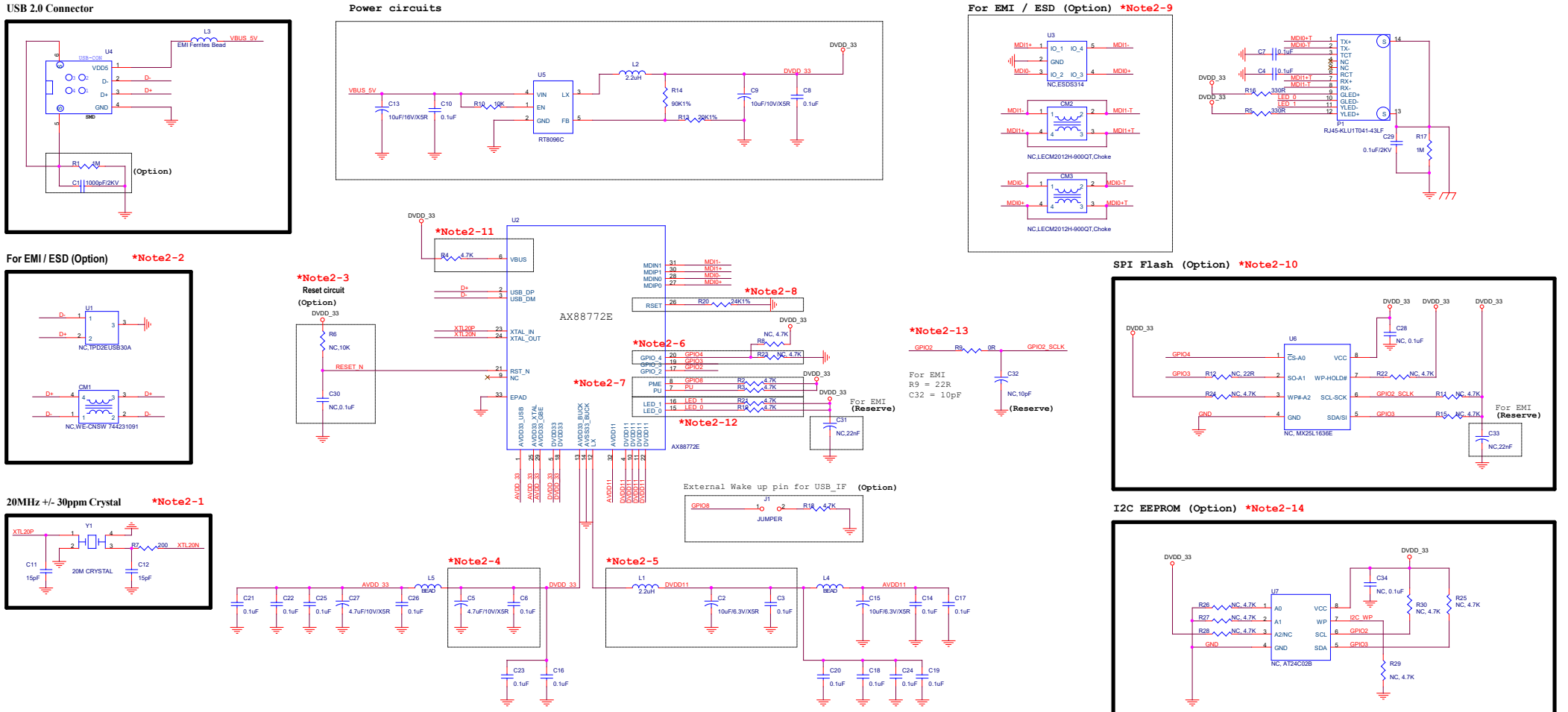


Note:
1.Please refer to AX88772E USB 2.0 to Fast Ethernet Application Design Note for more AX88772E PCB layout design notes.

2.Please deliver us your AX88772E schematic and PCB layout file for further review.

3.Please do full functionality testing on your own AX88772E board before running into production stage.

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*Note2-1:
The reference 20MHz crystal is the NSK
NXK20.000AC20F-BK4 SMD 20MHz crystal with CL 20pF and Drive Level 10uW. The 20MHz clock signals
should be within 20MHz +- 30ppm.
AX88772E does not support external oscillator clock source.

*Note2-2:
When you need to mount CM1 choke, please remember to cut the D+/D- traces below the CM1 choke.

*Note2-3:
The RC reset circuit is optional for AX88772E applications. You can reserve the RC reset circuit
on your AX88772E schematic to fine tune the reset timing if necessary.
For on-board self-power applications, designers can manually control the RESET_N signal via a
MCU's GPIO pin if necessary. Please reset AX88772E after the USB host/hub controllers are reset
completedly.

*Note2-4:
The capacitance C5, C6 must be closed to Pin13 (AVDD33_BUCK).

*Note2-5:
The capacitance C2, C3 and bead L1 must be closed to Pin12 (LX).

*Note2-6:
Please pull down GPIO4 to GND with 4.7k Ohm for AX88772E with I2C EEPROM (Option) if need.
The R9 and C32 must be close to AX88772E's GPIO2 pin.

*Note2-7:
PME/GPIO8 (pin8) must be alway be pulled high.
Please reserve the External WAKE circuit location if you need to run the USB-IF compliant test (mount
R18 4.7K resistor and mount a Button part at J1 location). Don't need mount J1, R18 in production.

*Note2-8:
The R20 resistor of RSET signal MUST be 24K 1%.

*Note2-9:
When you need to mount CM2/CM3 chokes, please remember to cut the MDI0+/MDI0-
/MDI1+/MDI1- traces below the CM2/CM3 chokes.
Please implement the ESD protection circuit based on the preferred TVS diode
part reference circuit.

*Note2-10:
The SPI Flash is optional for AX88772E.
External SPI flash component selection: The SCLK must be supported 40MHz for READ instruction and size
1Mbit or larger.
SPI flash must be close to AX88772E, keep the GPIO2 routing as short as possible.

*Note2-11:
The VBUS pin of AX88772E should be connected to the 3.3V level, which power source needs
from the VBUS signal(5V) of USB connector, to judge the USB port is attached or not.

*Note2-12:
AX88772E demo board's LED default setting:
LED_0 for Green LED Link (10M/100M)
LED_1 for Yellow LED Active (10M/100M)

*Note2-13:
The EMI reserved circuit is only for SPI Flash. When SPI Flash is not mounted,
please keep the EMI reserved circuit in NC.

*Note2-14:
The I2C EEPROM is optional for AX88772E.

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Revision History

Revision	Date	Comment
V1.00	2026/08/05	Initial release.
V1.01	2026/09/30	Added some description in Note 2-1.

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